

Cambridge International AS & A Level

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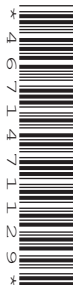
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CENTRE
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FURTHER MATHEMATICS

9231/21

Paper 2 Further Pure Mathematics 2

May/June 2021

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

- 1 (a) Given that a is an integer, show that the system of equations

$$ax + 3y + z = 14,$$

$$2x + y + 3z = 0,$$

$$-x + 2y - 5z = 17,$$

has a unique solution and interpret this situation geometrically.

[4]

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- (b) Find the value of a for which $x = 1$, $y = 4$, $z = -2$ is the solution to the system of equations in part (a).

[1]

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- 2** The variables x and y are related by the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 2x + 1.$$

- (a) Find the general solution for y in terms of x . [6]

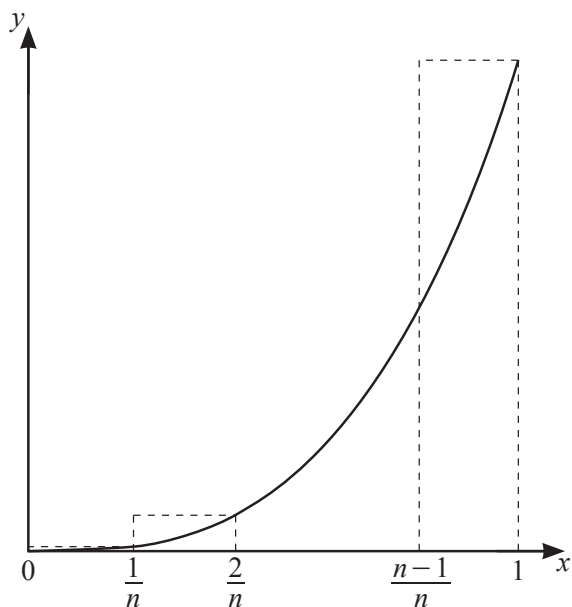
This image shows a full page of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.

- (b) State an approximate solution for large positive values of x . [1]

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The diagram shows the curve with equation $y = x^3$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

- (a) By considering the sum of the areas of these rectangles, show that $\int_0^1 x^3 dx < U_n$, where

$$U_n = \left(\frac{n+1}{2n}\right)^2. \quad [4]$$

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

- (b) Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 x^3 dx$. [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (c) Find the least value of n such that $U_n - L_n < 10^{-3}$. [2]

[illegible]

- 5** (a) State the sum of the series $z+z^2+z^3+\dots+z^n$, for $z \neq 1$. [1]

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- (b)** Given that z is an n th root of unity and $z \neq 1$, deduce that $1 + z + z^2 + \dots + z^{n-1} = 0$. [2]

[illegible]

- (c) Given instead that $z = \frac{1}{3}(\cos \theta + i \sin \theta)$, use de Moivre's theorem to show that

$$\sum_{m=1}^{\infty} 3^{-m} \cos m\theta = \frac{3 \cos \theta - 1}{10 - 6 \cos \theta} . \quad [7]$$

[illegible]

[illegible]

- 7 (a) It is given that $y = \operatorname{sech}^{-1}\left(x + \frac{1}{2}\right)$.

Express $\cosh y$ in terms of x and hence show that $\sinh y \frac{dy}{dx} = -\frac{1}{\left(x + \frac{1}{2}\right)^2}$. [3]

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- (b) Find the first three terms in the Maclaurin's series for $\operatorname{sech}^{-1}\left(x + \frac{1}{2}\right)$ in the form

$$\ln a + bx + cx^2,$$

where a , b and c are constants to be determined. [7]

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(a) Find $\frac{dx}{dt}$ and show that $\frac{dy}{dt} = 1 - \sinh^2 t$. [3]

(b) (i) Show that $A = \pi \int_0^1 \left(\frac{3}{2}t - \frac{1}{4} \sinh 2t \right) (1 + \cosh 2t) dt$. [4]

This image shows a full page of a worksheet designed for handwriting practice. It features 18 evenly spaced, horizontal dashed lines across the entire width of the page. The background is plain white, providing a clear guide for letter height and placement. There are no margins, text, or other markings present.

- (ii) Hence find A in terms of π , $\sinh 2$ and $\cosh 2$. [6]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dashed lines, creating a series of uniform gaps for letter height. The lines are evenly spaced across the entire page, providing a guide for consistent letter formation. There is no text or other markings on the page.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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